Math 74: Algebraic Topology

Prishita Dharampal

Credit Statement: Talked to Sair Shaikh'26

Problem 1.

(a) Let $p : (E, e_0) \to (B, b_0)$ be a covering map. Suppose that E is path-connected. Then show that there is a surjection,

$$\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$$

(b) Moreover, if E is simply connected, then ϕ is a bijection.

Hint: Try to modify the computation of $\pi_1(S^1, (1.0))$ for the first and second part.

Solution.

(a)

Problem 2. Let $P : E \to B$ be a covering map, with B connected. Then if $p^{-1}(b_0)$ has k-many elements fr some finite number k, $p^{-1}(b)$ also has k-many elements for every $b \in B$.

Problem 3. If $p : E \to B$ is a covering map with E path-connected and B simply-connected. Then ϕ is a homeomorphism.

Problem 4. If $p : E \to B$ and $p' : E' \to B'$ are covering maps, then so is $p \times p' : E \times E' \to B \times B'$, where $p \times p'$ is defined in an obvious manner.

Problem 5. Show that if a subspace A is a deformation retraction of a space X, and another subspace B is a deformation retraction of A, then B is a deformation retraction of X.

Problem 6. If a space retracts onto a subspace A, then the homomorphism on the fundamental group inducted by the inclusion $\iota : A \hookrightarrow X$ is injective. If A is a deformation retract of X, then $\iota_\star$ is an isomorphism.

Problem 7. Show that if a space X deformation retracts to a point $x \in X$, then for each neighborhood U of x in X there exists a neighborhood $V \subset U$ of x such that the inclusion map $V \hookrightarrow U$ is nullhomotopic (nullhomotopic means homotopic to a constant map).

Problem 8.

(a) Let X be the subspace of $\mathbb{R}^2$ consisting of the horizontal segment $[0, 1] \times 0$ together with all the vertical segments $r \times [0, 1 - r]$ for r a rational number in $[0, 1]$. Show that X deformation retracts to any point in the segment $[0, 1] \times 0$, but not to any other point. [See the preceding problem.]

1. Let Y be the subspace of $\mathbb{R}^2$ that is the union of an infinite number of copies of X arranged as in the figure below. Show that Y is contractible but does not deformation retract onto any point.

Problem 9. Define $f : S^1 \times I \to S^1 \times I$ by $f(\theta, s) = (\theta + 2\pi s, s)$, so f restricts to the identity on the two boundary circles of $S^1 \times I$. Show that f is homotopic to the identity by a homotopy f_t that is stationary on one of the boundary circles, but not by any homotopy f_t that is stationary on both boundary circles. [Consider what f does to the path $s \mapsto (\theta_0, s)$ for fixed $\theta_0 \in S^1$].